6. Distributions Derived from the Normal Distribution

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SS26 MTH344: Mathematical Statistics

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Lecture Topics

- When the sample size n is small, the CLT-based approach may have substantial approximation errors.
- This lecture reviews important distributions in finite sample statistical inference, with a particular focus on the case where $X_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with unknown parameters $(\mu, \sigma^2)^T$.
- See Chapter 6 in **Mathematical Statistics and Data Analysis [MSDA]** (Rice, 2007) for details. Selected examples and text excerpts are reproduced from [MSDA] for educational use only under fair use.
- This course presents content mostly following the textbooks, but also provides substantial unique and advanced material.

Introduction

- **Revisiting the Example of Testing the Fairness of a Coin**: Suppose we want to test if a coin is fair by flipping it n times. Let $X_i \overset{\text{i.i.d.}}{\sim} \mathrm{Ber}(0.5)$, and let $\bar{x}_n$ denote the observed sample mean. When $\bar{x}_n < 0.5$, a key quantity of interest is:

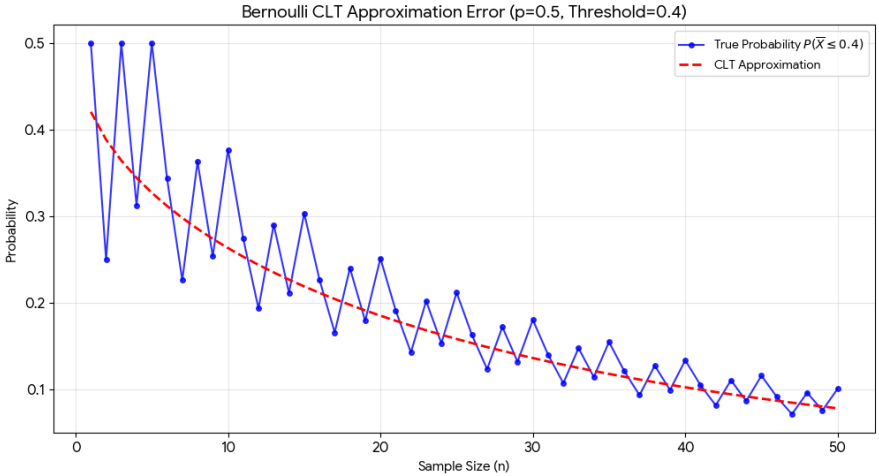
$$P(\bar{X}_n \leq \bar{x}_n). \tag{1}$$

- For $n = 100$ and $\bar{x}_{100} = 0.4$, the true probability based on the binomial PMF is 1.76%, whereas the CLT-based approximation:

$$P\left(\frac{\sqrt{n}\left(\bar{X}_n - 0.5\right)}{\sqrt{0.5(1-0.5)}} \leq \frac{\sqrt{n}\left(\bar{x}_n - 0.5\right)}{\sqrt{0.5(1-0.5)}}\right) \approx \Phi\left(\frac{\sqrt{n}\left(\bar{x}_n - 0.5\right)}{\sqrt{0.5(1-0.5)}}\right) \tag{2}$$

  yields 2.28%. Therefore, the approximation error is 0.58%.
- What if n is much smaller than 100?

Introduction



Introduction

- **Revisiting Example 5.3.E**: Let $X_1, X_2, \ldots, X_n$ be i.i.d. random variables with $\mathrm{E}[X_i] = \mu$ and $\mathrm{Var}[X_i] = \sigma^2$. They can be viewed as repeated measurements of a quantity μ. Using the CLT, we have:

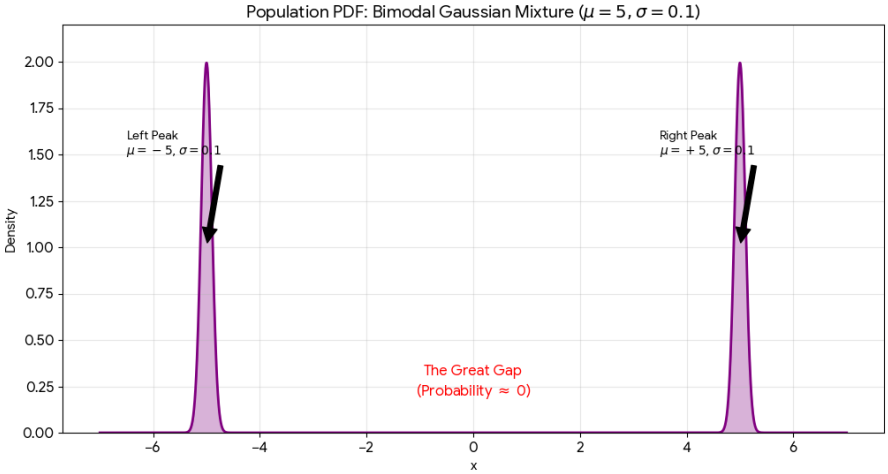
$$P(|\bar{X}_n - \mu| < c) \approx \Phi\left(\frac{c\sqrt{n}}{\sigma}\right) - \Phi\left(-\frac{c\sqrt{n}}{\sigma}\right) \tag{3}$$

- Let's examine the quality of this CLT-based approximation when the population consists of two heterogeneous groups:

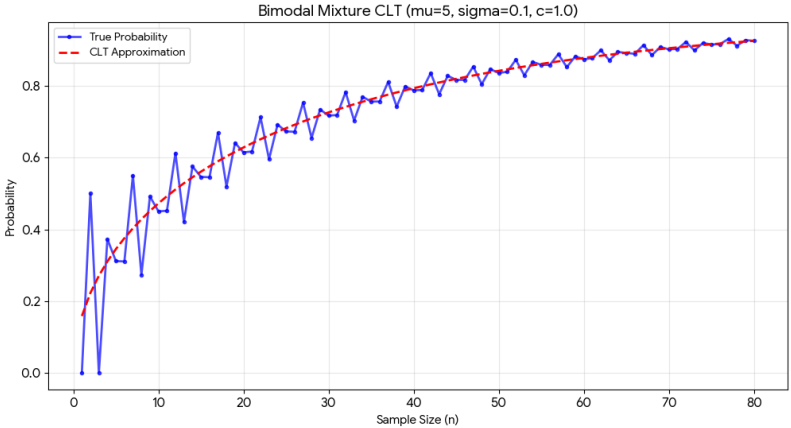
$$f(x) = \frac{1}{2}\mathcal{N}(x; -5, 0.1^2) + \frac{1}{2}\mathcal{N}(x; 5, 0.1^2). \tag{4}$$

  Here, $\mathrm{Var}[X_i] = \sigma^2 = 5^2 + 0.1^2$.
- We set $c = 1$ and compare the LHS and RHS in Equation (3).

Introduction



Introduction



Q: What is happening when $n = 3$?

Introduction

- Such approximation errors highlight the limitations of asymptotic theory in small samples.
- Finite sample inference addresses this challenge. These methods are broadly classified into two categories: (i) parametric and (ii) non-parametric.
- Let's begin with the simplest case where the population is distributed as $\mathcal{N}(\mu, \sigma^2)$ with unknown parameters $(\mu, \sigma^2)^T$. Note that:

$$Z := \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0,1) \quad \text{for any } n \in \mathbb{N}. \tag{5}$$

  Therefore, $F(z) = \Phi(z)$ where $z = \sqrt{n}(\bar{x}_n - \mu)/\sigma$.
- **Q**: For a given n and observed $\bar{x}_n$, can we directly use $\Phi(z)$ to examine how extreme the observed (standardized) sample mean is?

Introduction

- Suppose σ is unknown while μ is known. In such a case, a natural alternative is to replace σ with the sample standard deviation S_n, defining:

$$\frac{\bar{X}_n - \mu}{S_n/\sqrt{n}} \tag{6}$$

where $S_n = \sqrt{(n-1)^{-1}\sum_{i=1}^{n}\left(X_i - \bar{X}_n\right)^2}$.

- The *chi-square distribution* and the *t distribution* are key distributions used to describe the new random quantity in Equation (6).
- For an advanced analysis, we are often interested in comparing variances between two populations. In such a case, an important criterion is whether the ratio of sample variances from the two populations,

$$S_1^2/S_2^2, \tag{7}$$

is extremely large or small. The *F-distribution* relates to this quantity.

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χ^2 Distribution

- **Chi-square Distribution**: A random variable V is said to follow the *chi-square distribution with n degrees of freedom*, denoted by $\chi^2(n)$ where $n \in \mathbb{N}$, if there exist $Z_1, \ldots, Z_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1)$ such that:

$$V = Z_1^2 + Z_2^2 + \cdots + Z_n^2. \tag{8}$$

- The *degrees of freedom* of $V \sim \chi^2(n)$ refers to the number of independent standard Gaussian random variables squared and summed to construct V.
- $V = \boldsymbol{Z}^T I_n \boldsymbol{Z}$ where $\boldsymbol{Z} = (Z_1, Z_2, \ldots, Z_n)^T$. That is, V is a quadratic form in $\boldsymbol{Z}$ consisting of n i.i.d. standard Gaussian.

χ^2 Distribution

Exercise 1

Let $X \sim \mathcal{N}(\mu, \sigma^2)$. Show that,

$$V = \frac{(X-\mu)^2}{\sigma^2} \sim \mathrm{Gamma}\left(\frac{1}{2}, \frac{1}{2}\right). \tag{9}$$

χ^2 Distribution

Exercise 2

Show that the PDF of $\chi^2(n)$ is given by:

$$f(v) = \frac{1}{2^{n/2}\Gamma(n/2)} v^{(n/2)-1} e^{-v/2}, \quad v \geq 0. \tag{10}$$

χ^2 Distribution

Exercise 3

Show that the MGF of $\chi^2(n)$ is given by:

$$M(t) = (1 - 2t)^{-n/2}. \tag{11}$$

t Distribution

- <u>t **distribution**</u>: A random variable T is said to follow the *t distribution with n degrees of freedom*, denoted by $T \sim t(n)$, if there exist $Z \sim \mathcal{N}(0,1)$, $V \sim \chi^2(n)$, and $Z \perp\!\!\!\perp V$ such that:

$$T = \frac{Z}{\sqrt{V/n}}. \tag{12}$$

- Let $T \sim t(n)$. Since $E[|T|^k] = \infty$ when $k \geq n$, the MGF of $t(n)$ does not exist.

t Distribution

Exercise 4

Proposition 6.2.A: Show that the PDF of $t(n)$ is given by:

$$f(t) = \frac{\Gamma((n+1)/2)}{\sqrt{n\pi}\Gamma(n/2)} \left(1 + \frac{t^2}{n}\right)^{-(n+1)/2} \tag{13}$$

t Distribution

t Distribution

Exercise 5

Let $T \sim t(n)$. Show that $E[|T|^k] = \infty$ when $k \geq n$.

t Distribution

Exercise 6

Let $X_n \sim t(n)$. Show that $X_n \xrightarrow{d} \mathcal{N}(0, 1)$.

F Distribution

- <u>F-**distribution**</u>: A random variable W is said to follow the F *distribution with* n_1 *and* n_2 *degrees of freedom*, denoted by $W \sim F(n_1, n_2)$, if there exist $V_1 \sim \chi^2(n_1)$, $V_2 \sim \chi^2(n_2)$, and $V_1 \perp\!\!\!\perp V_2$ such that:

$$W = \frac{V_1/n_1}{V_2/n_2}. \tag{14}$$

- Let $W \sim F(n_1, n_2)$. Since $E[|W|^k] = \infty$ when $k \geq n_2/2$, the MGF of $F(n_1, n_2)$ does not exist.

F Distribution

Exercise 7

Proposition 6.2.B: Show that the PDF of $F(n_1, n_2)$ is given by:

$$f(w) = \frac{\Gamma((n_1+n_2)/2)}{\Gamma(n_1/2)\Gamma(n_2/2)} \left(\frac{n_1}{n_2}\right)^{n_1/2} w^{n_1/2-1} \left(1 + \frac{n_1}{n_2} w\right)^{-(n_1+n_2)/2}, \quad w > 0. \tag{15}$$

F Distribution

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The Sample Mean and the Sample Variance

- Let $X_1, \ldots, X_n$ be i.i.d. random variables (*samples*) from $\mathcal{N}(\mu, \sigma^2)$.
- Again, we denote the sample mean by:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i. \tag{16}$$

- We formulate the *sample variance* by:

$$S_n^2 = \frac{1}{n-1} \sum_{i=1}^{n} \left(X_i - \bar{X}_n \right)^2. \tag{17}$$

Note that $\mathrm{E}[S_n^2] = \sigma^2$.

The Sample Mean and the Sample Variance

Exercise 8

Let $\boldsymbol{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$. Show that for any real matrices A and B, if $\mathrm{Cov}[A\boldsymbol{X}, B\boldsymbol{X}] = 0$, then $A\boldsymbol{X} \perp\!\!\!\perp B\boldsymbol{X}$.

The Sample Mean and the Sample Variance

Exercise 9

Theorem 6.3.A: Let $X_1, \ldots, X_n$ be i.i.d. samples from $\mathcal{N}(\mu, \sigma^2)$. Show that $\bar{X}_n$ and $(X_1 - \bar{X}_n, \cdots, X_n - \bar{X}_n)^T$ are independent.

The Sample Mean and the Sample Variance

- **Corollary 6.3.A**: Let $X_1, \ldots, X_n$ be i.i.d. samples from $\mathcal{N}(\mu, \sigma^2)$, and $\bar{X}_n$ and S_n^2 be their sample mean and variance, respectively. Then,

$$\bar{X}_n \perp\!\!\!\perp S_n^2. \tag{18}$$

- **Theorem 6.3.B**: Let $X_1, \ldots, X_n$ be i.i.d. samples from $\mathcal{N}(\mu, \sigma^2)$, and S_n^2 be their sample variance. Then,

$$\frac{(n-1)S_n^2}{\sigma^2} \sim \chi^2(n-1). \tag{19}$$

- **Corollary 6.3.B**: Let $X_1, \ldots, X_n$ be i.i.d. samples from $\mathcal{N}(\mu, \sigma^2)$, and $\bar{X}_n$ and S_n^2 be their sample mean and variance, respectively. Then,

$$\frac{\bar{X}_n - \mu}{S_n/\sqrt{n}} \sim t(n-1). \tag{20}$$

The Sample Mean and the Sample Variance

Exercise 10

Let $\boldsymbol{Z} \sim \mathcal{N}(0_d, I_d)$ and let A be a symmetric idempotent matrix (i.e., $A^2 = A$ and $A^T = A$).[a] Show that:

$$\boldsymbol{Z}^T A \boldsymbol{Z} \sim \chi^2(\mathrm{tr}(A)). \tag{21}$$

[a] This condition is also necessary for the quadratic form to follow a chi-square distribution.

The Sample Mean and the Sample Variance

Exercise 11

Let $X_1, \ldots, X_n$ be i.i.d. samples from $\mathcal{N}(\mu, \sigma^2)$, and S_n^2 be their sample variance. Show that:

$$\frac{(n-1)S_n^2}{\sigma^2} \sim \chi^2(n-1). \tag{22}$$

The Sample Mean and the Sample Variance

- We can construct confidence intervals for μ and σ^2 based on the previous results.
- Let $t_\alpha(n)$ and $\chi^2_\alpha(n)$ be the upper α-th quantiles of $t(n)$ and $\chi^2(n)$, respectively, where $0 < \alpha < 0.5$. That is,

$$P(t_{1-\alpha/2}(n) \leq T \leq t_{\alpha/2}(n)) = P(\chi^2_{1-\alpha/2}(n) \leq V \leq \chi^2_{\alpha/2}(n)) = 1 - \alpha,$$

  where $T \sim t(n)$ and $V \sim \chi^2(n)$.[1]
- Then,

$$\begin{aligned} &P\left(t_{1-\alpha/2}(n-1) \leq \frac{\bar{X}_n - \mu}{S_n/\sqrt{n}} \leq t_{\alpha/2}(n-1)\right) \\ &= P\left(\bar{X}_n - t_{\alpha/2}(n-1)\frac{S_n}{\sqrt{n}} \leq \mu \leq \bar{X}_n + t_{\alpha/2}(n-1)\frac{S_n}{\sqrt{n}}\right) = 1 - \alpha. \end{aligned} \tag{23}$$

[1] Note that the direction of the upper quantile is opposite to that of the standard cumulative quantile.

The Sample Mean and the Sample Variance

- Similarly,

$$
\begin{aligned}
&P\left(\chi^2_{1-\alpha/2}(n-1) \le \frac{(n-1)S_n^2}{\sigma^2} \le \chi^2_{\alpha/2}(n-1)\right) \\
&= P\left(\frac{(n-1)S_n^2}{\chi^2_{\alpha/2}(n-1)} \le \sigma^2 \le \frac{(n-1)S_n^2}{\chi^2_{1-\alpha/2}(n-1)}\right) = 1-\alpha.
\end{aligned}
\tag{24}
$$

- Note that the interval endpoints depend only on observed data and not on unknown parameters.

The Sample Mean and the Sample Variance

- We can construct confidence intervals for the ratio of population variances from two independent normal populations.
- Let $X_1, \ldots, X_n$ be i.i.d. samples from $\mathcal{N}(\mu_X, \sigma_X^2)$ and $Y_1, \ldots, Y_m$ be i.i.d. samples from $\mathcal{N}(\mu_Y, \sigma_Y^2)$. Assume that the two samples are independent. Then, denoting the sample variances of X and Y by S_X^2 and S_Y^2, respectively:

$$\frac{S_X^2/\sigma_X^2}{S_Y^2/\sigma_Y^2} \sim F(n-1, m-1). \tag{25}$$

- Denoting the upper α-quantile of $F(n, m)$ by $F_\alpha(n, m)$, we have:

$$\begin{aligned} &P\left(F_{1-\alpha/2}(n-1, m-1) \leq \frac{S_X^2}{S_Y^2}\frac{\sigma_Y^2}{\sigma_X^2} \leq F_{\alpha/2}(n-1, m-1)\right) \\ &= P\left(\frac{S_X^2}{S_Y^2}\frac{1}{F_{\alpha/2}(n-1, m-1)} \leq \frac{\sigma_X^2}{\sigma_Y^2} \leq \frac{S_X^2}{S_Y^2}\frac{1}{F_{1-\alpha/2}(n-1, m-1)}\right) = 1-\alpha. \end{aligned} \tag{26}$$

Practice Problem I

1. Problem 7 in Chapter 6 in [MSDA].
2. Prove or provide a counterexample: Let $X \sim \chi^2(n)$ and $Y \sim \chi^2(n+1)$ for some $n \in \mathbb{N}$. Then, $\mathrm{Cov}[X, Y] = 2n$.
3. Prove or provide a counterexample: Let $X \sim \chi^2(n)$, $Y \sim \chi^2(m)$, and $X \perp\!\!\!\perp Y$. Then, $X + Y \sim \chi^2(n+m)$.

Advanced Practice Problem I

1. Let $Z \sim \mathcal{N}(0,1)$, $V \sim \chi^2(n)$, and $Z \perp\!\!\!\perp V$.
   a) Let $T = Z/\sqrt{V/n}$. Show that for $k < n$ and even k,

$$\mathrm{E}[T^k] = n^{k/2} \frac{\Gamma(\frac{k+1}{2})\Gamma(\frac{n-k}{2})}{\sqrt{\pi}\Gamma(\frac{n}{2})}. \tag{27}$$

   b) Let $R_{n,k} = \mathrm{E}[T^k] - \mathrm{E}[Z^k]$. Show that for any $0 < \epsilon < 1$ and even k,

$$\lim_{n\to\infty} n^{\epsilon} R_{n,k} = 0. \tag{28}$$

References I

Rice, J. A. (2007). *Mathematical statistics and data analysis*, volume 371. Thomson/Brooks/Cole Belmont, CA.